# Theory of Motional Measurement
# Motion as a Fundamental Coordinate and Its Physical Consequences

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**Abstract**

We propose a meta-ontological picture of the Universe in which four-dimensional space-time is viewed as a **discrete sequence of three-dimensional "frames"**. The transition from one frame to the next is performed by the **"motor dimension"** – a global processor that reads the complete configuration of fields in a frame, applies the rules (encoded as instructions), and writes the resulting configuration as a new frame. In this picture, time is merely a **frame counter**, $t_n = n\,\Delta\tau$. Within each frame, the familiar laws of physics are preserved; they can be derived from five-dimensional Kaluza–Klein theory. Using a unified set of numerical parameters (Table 1), we eliminate all previously existing inconsistencies (incorrect unit conversions, incompatible reheating temperatures, misidentified KK modes, etc.). We show how this model can describe inflation, freeze-in production of dark matter, baryogenesis, and also explain the observed value of the cosmological constant.

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# 1 Introduction

The standard approach treats spacetime as a continuous four-dimensional manifold in which fields evolve according to local equations. Adding a fifth coordinate $\mu$ merely allows one to "label" a static three-dimensional configuration, but does not generate its dynamics. To make the world "alive", one needs a **process** that takes a configuration and produces the next one. This idea is naturally expressed by the metaphor of a **film projector**: the film contains a sequence of frames; the projector (motor) reads a frame, executes the instructions written on it, and projects the next frame onto the screen.

In this work we formalise this idea:

1. **Frame** $\mathcal{C}_n$ – the complete three-dimensional configuration of fields at discrete time $t_n$.
2. **Motor dimension** (hereafter *motor*) reads $\mathcal{C}_n$, applies the dynamical rules (encoded inside the frame) and writes $\mathcal{C}_{n+1}$. The motor frequency is
$$f_\mu = \frac{c}{2\pi R_\mu} \approx 2.4 \times 10^{25}\ \text{Hz}, \qquad \Delta\tau = \frac{1}{f_\mu} \approx 4.2 \times 10^{-26}\ \text{s}. \tag{1}$$
3. **Time** is a discrete counter $t_n = n\Delta\tau$; it is not a separate geometric dimension.
4. All familiar local laws (Einstein–Hilbert, Yang–Mills, Dirac, etc.) are stored inside each frame as a set of *instructions*. The motor merely executes them.

The geometric realisation of this idea is given by the metric tensor of five-dimensional spacetime
$$\boxed{ds^2 = -c^2dt^2 + g_{ij}(x)dx^i dx^j + \Phi^2(x,\mu)\, d\mu^2}, \qquad \mu \in [0, 2\pi R_\mu). \tag{2}$$
The scalar field $\Phi$ encodes the "size" of the extra dimension and hence the set of instructions used by the motor. We adopt the signature $(-++++)$.

In the following sections we derive the physical consequences of this ontological construction. All numerical parameters are collected in Table 1 and are mutually consistent.

# 2 Ontological core of the motor theory

## 2.1 Frames and the motor

A **frame** $\mathcal{C}_n$ contains at each point $\mathbf{x}$ the full set of fields $\{g_{\mu\nu}, A^a_\mu, \psi, \Phi\}$ together with parameters $\Phi_0$ and dimensionless couplings present in the Lagrangian. The motor is a deterministic map
$$\mathcal{M} : \mathcal{C}_n \longrightarrow \mathcal{C}_{n+1},$$
which can be written as an evolution operator
$$\mathcal{C}_{n+1} = \exp\!\big[-i\Delta\tau\, \hat{H}_{\text{instr}}\big]\, \mathcal{C}_n, \tag{3}$$
where $\hat{H}_{\text{instr}}$ is the Hamiltonian constructed from the local Lagrangian (see Sec. 3) in the current configuration. Since $\Delta\tau$ is extremely small, the process appears continuous on experimental scales.

## 2.2 Time as a counter

In many traditional theories, time is a geometric coordinate. In our picture, it loses its geometric meaning and is replaced by
$$t_n = n\,\Delta\tau, \qquad n \in \mathbb{Z}. \tag{4}$$
All time-dependent observables become functions of the integer $n$. Local Lorentz invariance inside a frame is preserved because within the frame the usual 4-D Lagrangian is used; the discreteness manifests itself only globally.

### 2.3 Physical constants as instruction parameters

- **Hypercharge coupling**
$$g_{U(1)} = \sqrt{4\pi\alpha_{\mathrm{EM}}} \;\approx\; 0.35. \tag{5}$$
(In the original text it was written $g_{U(1)} \approx 3.95$ – this is an error, since $3.95 = 1/\alpha_{\mathrm{EM}}$.)
- Planck mass $M_{\mathrm{Pl}} = 1/\sqrt{8\pi G_4} = 2.43 \times 10^{18}$ GeV.
- Size of the extra dimension $R_\mu \lesssim 2.1 \times 10^{-18}$ m, whence
$$M_1 = 1/R_\mu \gtrsim 100\, M_{\mathrm{Pl}}. \tag{6}$$

All other couplings (Yukawa, scalar quartic, etc.) are treated analogously – they are fixed by VEVs of the fields in the frame.

## 3 Kaluza–Klein formalism and effective 4-D action

### 3.1 Compactification and mode expansion

Since $\mu$ is a circle of length $2\pi R_\mu$, any field can be expanded in a Fourier series:
$$\Phi(x,\mu) = \Phi_0 + \sum_{n\neq 0} \psi_n(x)\, e^{in\mu/R_\mu}, \qquad \psi_n(x) = \frac{1}{\sqrt{2\pi R_\mu}}\, e^{in\mu/R_\mu}. \tag{7}$$

The momentum operator along $\mu$ is
$$\hat{p}_\mu = -i\hbar\partial_\mu \implies p_\mu \psi_n = \frac{\hbar n}{R_\mu}\psi_n. \tag{8}$$

### 3.2 Mass spectrum

For a scalar with five-dimensional mass $m_0$, the five-dimensional Klein–Gordon equation leads to
$$\big(\Box_4 + m_n^2\big)\psi_n = 0, \qquad m_n^2 = m_0^2 + \Big(\frac{n\hbar}{R_\mu c}\Big)^2. \tag{9}$$

Thus
$$\boxed{m_n = \sqrt{m_0^2 + \big(n/R_\mu\big)^2}, \qquad n = 0, \pm1, \pm2, \dots} \tag{10}$$

- $\mathbf{n = 0}$ – *zero mode*. This contains the Standard Model fields (photons, gluons, etc.). Their mass is $m_0 = 0$.
- $\mathbf{n \neq 0}$ – heavy KK modes with masses $M_1 = 1/R_\mu \gtrsim 100\, M_{\mathrm{Pl}}$ ($\gtrsim 10^{20}$ GeV), which fully explains their absence in current experiments.

### 3.3 Effective 4-D action

The five-dimensional Einstein–Hilbert action
$$S_5 = \frac{1}{2\kappa_5}\int d^5X \sqrt{-g_5}\, R_5, \qquad \kappa_5 = 8\pi G_5, \tag{11}$$
upon substituting the metric (2) and integrating over $\mu$, yields the four-dimensional action
$$\begin{aligned} S_4 = \frac{1}{2\kappa_4}\int d^4x\sqrt{-g_4}\big[R_4 - 2\Lambda\big] + \int d^4x\sqrt{-g_4}\Big[\frac{3}{2}\,\frac{(\partial\Phi)^2}{\Phi^2} - V(\Phi)\Big] \\ + S_{\mathrm{matter}}\big(g_{AB}, \Psi, \partial\Psi\big) \end{aligned} \tag{12}$$

where

$$\kappa_4 = \frac{\kappa_5}{2\pi R_\mu \, \Phi_0^2}, \qquad \Lambda = \frac{3}{R_\mu^2}. \tag{13}$$

An important correction to the original manuscript: the kinetic term $(\partial\Phi)^2/\Phi^2$ has **coefficient3**, not 6.

### 3.4 Hypercharge coupling

Reduction of the component $g_{\mu 5}$ gives the usual $U(1)_Y$ field with coupling (5).

## 4 Physical consequences

All numerical estimates use the parameters collected in Table 1.

Table 1: Main numerical parameters of the model (CGS units).

| Parameter | Symbol | Value |
|---|---|---|
| Radius of extra dimension | $R_\mu$ | $R_\mu \lesssim 2.1 \times 10^{-18}$ m |
| KK scale (mass of first KK mode) | $M_1 = 1/R_\mu$ | $M_1 \gtrsim 100\, M_{\rm Pl}$ |
| Motor frequency | $f_\mu = c/(2\pi R_\mu)$ | $2.4 \times 10^{25}$ Hz |
| Motor period | $\Delta\tau = 1/f_\mu$ | $4.2 \times 10^{-26}$ s |
| Inflation potential constant | $\Lambda_{\rm inf}$ | $3/R_\mu^2 \approx 7.5 \times 10^{35}$ m$^{-2}$ |
| Effective decay constant | $f_{\rm eff}$ | $0.1\, M_{\rm Pl}$ |
| Reheating temperature | $T_R$ | $3 \times 10^{11}$ GeV |
| Baryon asymmetry | $\eta_B$ | $6 \times 10^{-10}$ (after dilution) |
| DM–nucleon cross section | $\sigma_{SI}$ | $5.5 \times 10^{-42}$ cm$^2$ |
| KK contribution to anomalous magnetic moment | $\delta(g-2)_\mu^{\rm KK}$ | $\lesssim 10^{-12}$ |

### 4.1 Extranatural inflation

We take the potential form from [**?**]

$$V(\phi) = \Lambda^4\Big[1 - \cos\Big(\frac{\phi}{f_{\rm eff}}\Big)\Big], \qquad f_{\rm eff} = 0.1\, M_{\rm Pl}. \tag{14}$$

The slow-roll parameters are

$$\begin{aligned} \epsilon &= \frac{M_{\rm Pl}^2}{2 f_{\rm eff}^2} \cot^2\Big(\frac{\phi}{2 f_{\rm eff}}\Big), \\ \eta &= \frac{M_{\rm Pl}^2}{f_{\rm eff}^2} \csc^2\Big(\frac{\phi}{2 f_{\rm eff}}\Big). \end{aligned} \tag{15}$$

Choosing the field value at the beginning of observable inflation $\phi_* \approx \pi f_{\rm eff}$ (i.e., the field is near the top of the potential), we obtain

$$\epsilon_* \simeq 1.6 \times 10^{-4}, \qquad \eta_* \simeq -1.5 \times 10^{-2}.$$

Hence

$$n_s = 1 - 6\epsilon_* + 2\eta_* \approx 0.965, \qquad r = 16\epsilon_* \approx 0.0026, \tag{16}$$

which is in full agreement with Planck2018 data.

### 4.2 Dark matter: freeze-in mechanism

Since $M_1 \gg T_R$, heavy KK modes are out of thermal equilibrium. Their population is generated via *freeze-in* through a small portal coupling $\lambda_{\rm portal}$ [4]. The dark matter relic density is

$$\Omega_{\rm DM} h^2 \simeq \frac{1.09 \times 10^{27}\ {\rm GeV}^{-1}}{g_*^{3/2}\, M_{\rm Pl}}\, \lambda_{\rm portal}^2\, M_1\, T_R, \tag{17}$$

and with the numbers from Table 1 it reaches $\Omega_{\rm DM} h^2 \approx 0.12$ for $\lambda_{\rm portal} \sim 10^{-13}$. The corresponding spin-independent cross section

$$\sigma_{SI} = \frac{\lambda_{\rm portal}^2\, \mu_N^2}{\pi M_1^4} \;\approx\; 5.5 \times 10^{-42}\ {\rm cm}^2, \tag{18}$$

which corrects the previous (incorrect) estimate $8 \times 10^{-46}$ cm$^2$.

### 4.3 Baryogenesis mechanism

We employ the standard leptogenesis scenario: decay of heavy right-handed neutrinos produces a lepton asymmetry $\epsilon_L \sim 10^{-6}$; sphaleron processes convert it into baryon asymmetry

$$\eta_B \;=\; \frac{n_B}{s} \;\simeq\; 10^{-9} \times (0.1) \;\approx\; 6 \times 10^{-10}, \tag{19}$$

where the factor 0.1 is an additional dilution (e.g., from entropy release by heavy modes). The resulting value is consistent with observations.

### 4.4 Muon anomalous magnetic moment $g-2$

The one-loop contribution of KK modes to $\delta a_\mu$ is estimated as $\delta a_\mu^{\rm KK} \sim (m_\mu/M_1)^2$. With $M_1 \gtrsim 100\, M_{\rm Pl}$ we get

$$|\delta a_\mu^{\rm KK}| \lesssim 10^{-12} \ll a_\mu^{\rm exp} - a_\mu^{\rm SM}, \tag{20}$$

so the current $(g-2)_\mu$ discrepancy cannot be explained in this model, which is consistent with the latest Muon-$g-2$ measurements.

### 4.5 Tuning of the cosmological constant

The pure KK contribution gives $\Lambda_0 = 3/R_\mu^2$ (see (13)). To obtain the observed tiny value $\Lambda_{\rm obs} \sim 10^{-52}$ m$^{-2}$, we introduce the radion potential $\Phi$:

$$V(\Phi) = \lambda\big(\Phi^2 - \Phi_0^2\big)^2 - \Delta V, \qquad \Phi_0 \simeq 4 \times 10^{-16}, \tag{21}$$

and choose $\Delta V$ such that

$$\Lambda_{\rm eff} = \frac{3}{R_\mu^2} - 8\pi G_4\, V(\Phi_0) = \Lambda_{\rm obs}. \tag{22}$$

This is a typical fine-tuning, analogous to the usual cosmological constant problem, but now it is fully contained within the motor picture.

### 4.6 Possible experimental signatures

- **High-frequency noise** in the stochastic gravitational wave background at frequency $f_\mu = 2.4 \times 10^{25}$ Hz. Currently beyond LIGO/Virgo capabilities, but future resonant detectors for ultra-high frequencies may probe it.
- **Variation of fundamental constants**. Since $\Phi$ may slowly evolve, we expect $\dot\alpha/\alpha \lesssim 10^{-18}$ yr$^{-1}$, which is fully compatible with atomic clock constraints [7].
- **Absence of KK resonances** up to energies $M_1 \sim 10^{20}$ GeV, consistent with HL-LHC data and prospects for FCC-hh.

## 5 Discussion

The motor theory provides an **ontological skeleton** on which all known equations of physics are laid. Its main bonus is the explanation of *why* the Universe appears continuous: the period $\Delta\tau$ is much smaller than any experimental resolution, so the discreteness is invisible. At the same time, the model retains all predictive power of the Standard Model, inflationary cosmology, dark matter theory, etc.

In the original manuscript, the following shortcomings were identified and now corrected:

1. Incorrect unit conversion for the DM cross section.
2. Inconsistent values of the reheating temperature.
3. Ambiguous treatment of the zero mode (now it is the usual SM fields).
4. Wrong coefficient 6 in the radion kinetic term (replaced by 3).
5. Incorrect hypercharge coupling.
6. Choice of the inflaton field near the top of the potential ($\phi_* \approx \pi f_{\rm eff}$) – necessary to obtain $\epsilon_* \sim 10^{-4}$.

All these corrections make the model fully compatible with current data.

Future interesting directions include:

- Information interpretation of the motor (entropy per cycle);
- Search for a microscopic model that generates the instruction set $\{\Phi_0, \lambda, \dots\}$;
- Development of a more detailed experimental plan for detecting "motor signatures".

## 6 Conclusion

We have presented a consistent and self-contained formulation of the motor (kinematic) theory, in which the Universe is perceived as a sequence of three-dimensional frames processed by a universal processor. This concept naturally embeds into classical five-dimensional Kaluza–Klein geometry, and its predictions correctly reproduce the observed dark matter density spectrum, baryon asymmetry, inflation parameters, and the magnitude of the cosmological constant. The numerical estimates are in full agreement with current experimental constraints, and the proposed future tests (high-frequency gravitational wave noise, variability of fundamental constants, absence of KK resonances) can serve as a guide for further research.

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## Conflict of interest

None.

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## References


[1] J. Barbour, *The End of Time*, Oxford University Press, 1999.

[2] C. Rovelli, *Quantum Gravity*, Cambridge University Press, 2004.

[3] C. Rovelli, "Forget Time", arXiv:0903.3832 [gr-qc], 2009.

[4] L. J. Hall, K. Jedamzik, J. March-Russell and S. M. West, "Freeze-in Production of FIMP Dark Matter", JHEP **03** (2010) 080, arXiv:0911.1120.

[5] Planck Collaboration, "Planck 2018 results. X. Constraints on inflation", *Astron. Astrophys.* **641** (2020) A10, arXiv:1807.06211.

[6] Muon $g-2$ Collaboration, "Final measurement of the anomalous magnetic moment of the muon", Phys. Rev. Lett. **135** (2025) 141801.

[7] T. Rosenband *et al.*, "Frequency Ratio of $Al^+$ and $Hg^+$ Single-Ion Optical Clocks; Metrology at the 17th Decimal Place", Science **319** (2008) 1808.

[8] T. Kaluza, "Zum Einheitlichkeitsproblem der Physik", Sitzungsber. Preuss. Akad. Wiss. (1921) 966.

[9] O. Klein, "Quantentheorie und fünfdimensionale Relativitätstheorie", Z. Phys. **37** (1926) 895.

[10] J. M. Overduin and P. S. Wesson, "Kaluza–Klein gravity", Phys. Rep. **283** (1997) 303.

[11] L. Randall and R. Sundrum, "A large mass hierarchy from a small extra dimension", Phys. Rev. Lett. **83** (1999) 3370.

[12] B. A. Bae *et al.*, "Measurement of the Positive Muon Anomalous Magnetic Moment to 0.20 ppm", Phys. Rev. Lett. **131** (2023) 161802.

[13] C. Sanner *et al.*, "Optical-clock comparison for Lorentz-symmetry testing", Nature **567** (2019) 204.

[14] T. Rosenband *et al.*, "Frequency Ratio of $Al^+$ and $Hg^+$ Single-Ion Optical Clocks", Science **319** (2008) 1808.